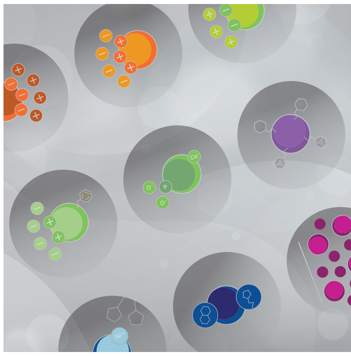
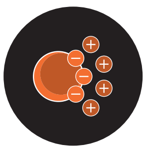
Chromatography



Nuvia HR-S Cation Exchange Resin



CATION EXCHANGE

- Best-in-class resolution
- High recovery of target biomolecules
- Robust polymeric matrix
- Scalable to industrial purification applications
- Full regulatory support

Exceptional Resolution of Related Species for Bioprocessing

Nuvia HR-S Resin is a strong cation exchanger (CEX), optimized for particle size and chemistry, that provides exceptional resolution and high recovery. The resin demonstrates fast mass transfer kinetics and superlative flow characteristics, chemical stability, and scalability. Nuvia HR-S Resin is the preferred solution when process developers face challenging separations of closely related biomolecules.

High Resolution

Improvements in upstream cell culture expression systems have produced high feed stream titers that place a larger demand on downstream purifications to separate closely related contaminants from target biomolecules. In monoclonal antibody therapeutics, high molecular weight impurities in the form of aggregates are a key challenge for process developers. Nuvia HR-S Resin delivers the resolution performance that bioprocess scientists need to address this challenge. The technical properties of Nuvia HR-S Resin are listed in Table 1.

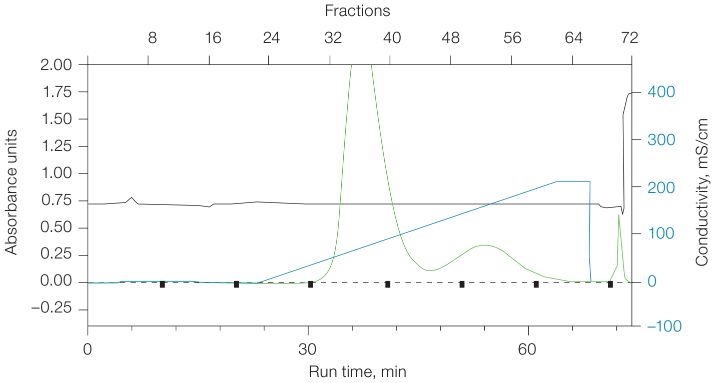
An example of a high-resolution separation is shown in Figure 1, where a mixture of monomer and aggregate was separated on Nuvia HR-S Resin. From a starting aggregate concentration of 8.9%, the final eluate pool contained 0.11% residual aggregate with an overall monomer recovery of 82%. Furthermore, as shown in Figure 2, better resolution was obtained using the Nuvia HR-S Resin with near baseline resolution of a protein mixture compared to Resin 1, another commercially available high-resolution small-particle cation exchanger.

Table 1. Properties of Nuvia HR-S Resin.

Property	Description
Type of ion exchanger	Strong cation
Functional group	$-SO_3^-$
Base matrix composition	Polyacrylamide
Mean particle size	50 µm
Total ionic capacity	100–180 µeq/ml
Dynamic binding capacity*	>70 mg/ml at 300 cm/hr
Recommended linear flow rate	50–200 cm/hr
Compression factor (settled bed volume/packed bed volume)	1.20–1.30
pH stability	Short term: 2–14 Long term: 4–13
Shipping solution	20% ethanol + 0.1 M NaCl or 2% benzyl alcohol
Regeneration	1–2 M NaCl
Sanitization	0.5–1.0 M NaOH
Storage conditions	20% ethanol or 0.1 M NaOH
Storage temperature	Room temperature
Chemical stability	1.0 M NaOH (20°C), up to 5 weeks or 0.1 M NaOH (20°C), up to 5 years
Shelf life	5 years

* 10% breakthrough capacity determined with 5.0 mg/ml human IgG in 20 mM Na acetate, pH 5.0.

A



B

Chromatography Condition	Specification
Column	7 x 56 mm Nuvia HR-S Resin packed bed column
Load	20 ml aggregate and monomer mixture at a loading level of 46 mg/ml
Flow rate	150 cm/hr
Equilibration	40 mM Na acetate, pH 5.0
Gradient elution	40 mM Na acetate, 0–0.4 M NaCl, pH 5.0

Fig. 1. Monoclonal antibody separation on Nuvia HR-S Resin. A, chromatogram of the purification of monomer from aggregate; **B**, chromatography conditions. A_{280} (—); pH (—); gradient elution (—).

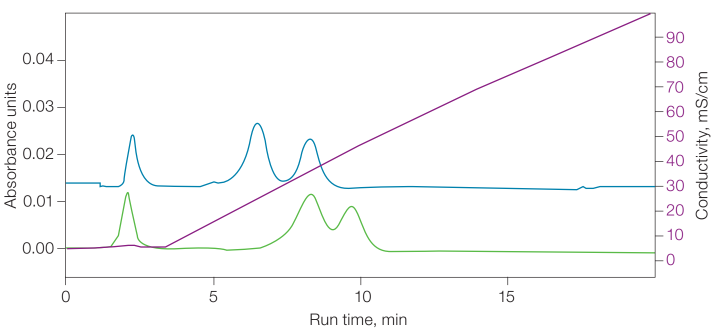


Fig. 2. Resolution performance of Nuvia HR-S Resin. Comparison of Nuvia HR-S Resin to another commercially available small-particle cation exchanger (Resin 1). Column size, 7 x 56 mm. A protein standard mixture of myoglobin, ribonuclease, and cytochrome C was loaded onto the columns in 20 mM sodium phosphate, pH 7.0, washed, and then eluted with a gradient of 20 mM sodium phosphate, pH 7.0 + 1 M NaCl. Nuvia HR-S (—); Resin 1 (—).

Consistent and Reproducible Performance

Chromatography media and resins used in downstream purification are required to perform consistently, even after repeated exposure to caustic sanitization protocols. Nuvia HR-S Resin binding capacity and protein recovery remained unchanged after exposure to 840 hr of 1.0 N NaOH (equivalent to 420 cycles at 2-hr sanitization hold), thus ensuring the resin can be used repeatedly without compromising performance, as shown in Figure 3.

Produced in a validated manufacturing process under strict specifications, the cornerstone of Nuvia HR-S Resin is batch-to-batch reproducibility. With more than 50 years of experience in chromatography, Bio-Rad understands and is committed to maintaining the highest level of quality and security of supply.

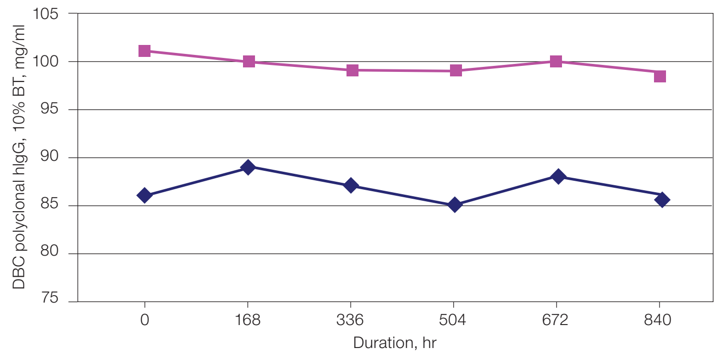


Fig. 3. Stability of Nuvia HR-S Resin. Results of hIgG binding capacity after exposing Nuvia HR-S Resin to 1.0 N NaOH. 10% breakthrough (—); % recovery (—). BT, breakthrough; DBC, dynamic binding capacity.

Comparative Data

Superior Aggregate Content and Monomer Recovery Using Nuvia HR-S

Nuvia HR-S Resin was compared to a commercially available small particle size CEX resin (Resin 1) to determine the aggregate content and monomer recovery as fractions were progressively pooled during elution. Nuvia HR-S Resin delivered a final aggregate content of <0.3% and a recovery of >80%, while the recovery using agarose-based Resin 1 was <70% for the same final aggregate content (Figure 4).

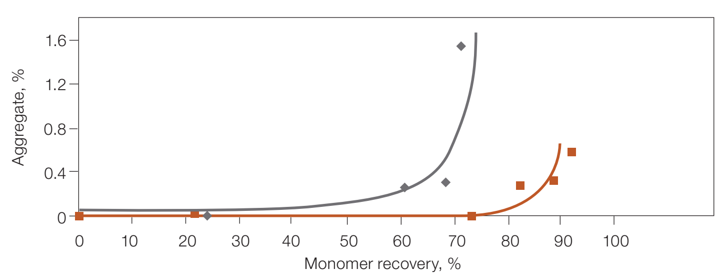


Fig. 4. Performance of Nuvia HR-S (■) vs. Resin 1 (◆).

Minimal On-Column Aggregate Generation

Six commercial resins including Nuvia HR-S were tested for the formation of on-column aggregates during mAb purification. Nuvia HR-S produced the smallest amount of on-column aggregates, as shown in Figure 5.

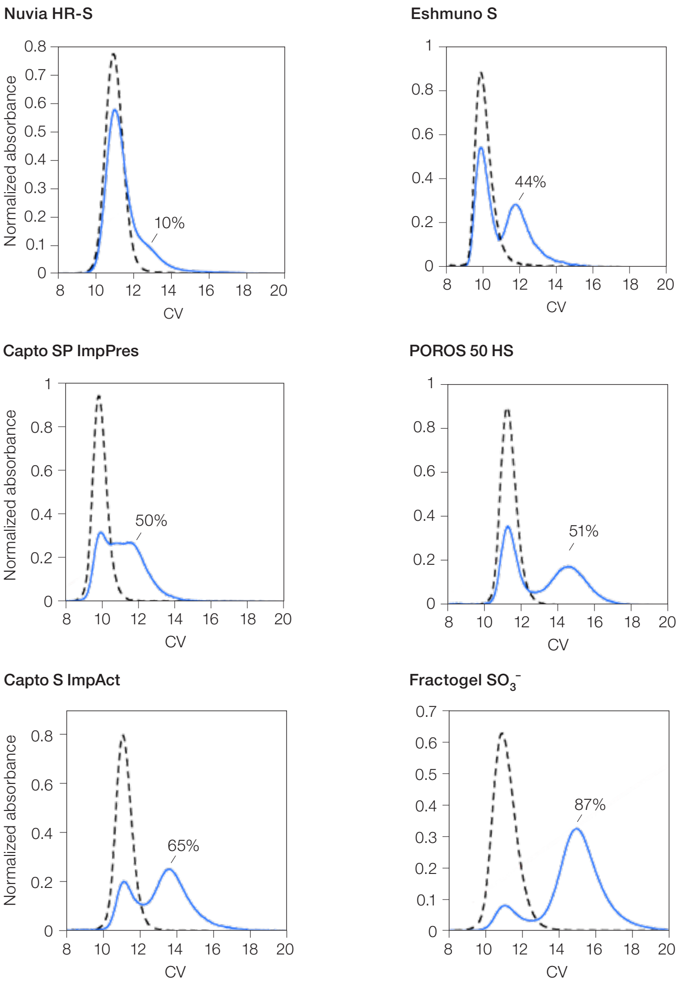


Fig. 5. On-column aggregates. 0 min hold time (– – –); 1,000 min hold time (——); percentage of on-column aggregate shown in plots. For further details, refer to Bio-Rad bulletin 7000, Minimizing On-Column Monoclonal Antibody Aggregate Formation.

Viral Clearance

Nuvia HR-S viral clearance was evaluated using xenotropic murine leukemia virus (XMuLV), an enveloped mRNA retrovirus. Studies were conducted with a third party in accordance with ICH Q5A(R2)* guidelines. Purified IgG monoclonal antibody (mAb) was prepared in 20 mM Na acetate, 25 mM NaCl, pH 5.0 to a concentration of 5 mg/ml. The samples were spiked with virus stock solution and loaded at >50 mg protein/ml of resin onto a prepacked 5 ml Foresight™ Nuvia HR-S Column (Bio-Rad, catalog #7324743). The mAb was eluted using ten column volumes (CV) at 300 cm/hr with a 0–100% linear gradient of 20 mM acetate, 500 mM NaCl, pH 5.0. The $\log_{10}$ reduction factor was 3.76 logs.

* International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use guidance entitled Q5A(R2) Viral Safety Evaluation of Biotechnology Products Derived from Cell Lines of Human or Animal Origin.

Ordering Information

Catalog #	Description
Bulk Resins	
1560511	**Nuvia HR-S Media**, 25 ml
12003712	**Nuvia HR-S Media (in BA)**, 25 ml
1560513	**Nuvia HR-S Media**, 100 ml
12003713	**Nuvia HR-S Media (in BA)**, 100 ml
156-0515	**Nuvia HR-S Media**, 500 ml
12003714	**Nuvia HR-S Media (in BA)**, 500 ml
12018125	**Nuvia HR-S Media**, 5 L
156-0517	**Nuvia HR-S Media**, 10 L
12003715	**Nuvia HR-S Media (in BA)**, 10 L

BA, 2% benzyl alcohol.

Catalog #	Description
Prepacked Screening Tools	
12009310	**EconoFit Cationic IEX Kit**, includes 1 ml each of Macro-Prep™ High S, UNOsphere S, Nuvia S, Nuvia HR-S, and Macro-Prep CM Resins
12009284	**EconoFit Nuvia HR-S Column**, 1 x 1 ml, 7 x 25 mm
732-4707	**Foresight Nuvia HR-S Plates**, 2 x 96-well, 20 µl
732-4831	**Foresight Nuvia HR-S RoboColumn Unit**, 200 µl
732-4832	**Foresight Nuvia HR-S RoboColumn Unit**, 600 µl
732-4723	**Foresight Nuvia HR-S Column**, 1 ml
732-4743	**Foresight Nuvia HR-S Column**, 5 ml

Technical Assistance

A regulatory support file is available upon request. Bio-Rad Laboratories, Inc. is an ISO 13485 registered corporation.

Visit **bio-rad.com/NuviaHRS** for more information and to request samples.

Visit **bio-rad.com/ProcessResins** for more information about Bio-Rad's complete line of process chromatography supports.

For additional information and technical assistance, contact your local Bio-Rad office or email our process specialists at **process@bio-rad.com.** In the U.S. and Canada, call 1-800-4BIORAD (1-800-424-6723).

Bio-Rad Laboratories, Inc.

Life Science Group

Website *bio-rad.com* ***USA*** *1 800 424 6723* ***Australia*** *61 2 9914 2800* ***Austria*** *00 800 00 24 67 23* ***Belgium*** *00 800 00 24 67 23* ***Brazil*** *4003 0399*
Canada *1 905 364 3435* ***China*** *86 21 6169 8500* ***Czech Republic*** *00 800 00 24 67 23* ***Denmark*** *00 800 00 24 67 23* ***Finland*** *00 800 00 24 67 23*
France *00 800 00 24 67 23* ***Germany*** *00 800 00 24 67 23* ***Hong Kong*** *852 2789 3300* ***Hungary*** *00 800 00 24 67 23* ***India*** *91 124 4029300* ***Israel*** *0 3 9636050*
Italy *00 800 00 24 67 23* ***Japan*** *81 3 6361 7000* ***Korea*** *82 080 007 7373* ***Luxembourg*** *00 800 00 24 67 23* ***Mexico*** *52 555 488 7670*
The Netherlands *00 800 00 24 67 23* ***New Zealand*** *64 9 415 2280* ***Norway*** *00 800 00 24 67 23* ***Poland*** *00 800 00 24 67 23* ***Portugal*** *00 800 00 24 67 23*
Russian Federation *00 800 00 24 67 23* ***Singapore*** *65 6415 3188* ***South Africa*** *00 800 00 24 67 23* ***Spain*** *00 800 00 24 67 23* ***Sweden*** *00 800 00 24 67 23*
Switzerland *00 800 00 24 67 23* ***Taiwan*** *886 2 2578 7189* ***Thailand*** *66 2 651 8311* ***United Arab Emirates*** *36 1 459 6150* ***United Kingdom*** *00 800 00 24 67 23*

Bulletin 6448 Ver B US/EG

24-0489 0824 Sig 0124